



UNIVERSITY OF GHANA

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DEPARTMENT OF STATISTICS AND ACTUARIAL SCIENCE

INTERIM ASSESSMENT II : 2018/2019

STAT 112: ELEMENTARY PROBABILITY (3 CREDITS)

INSTRUCTION: Answer all questions from Section A and any one from Section B.

DURATION: 1hr : 30 Minutes

SECTION A (40 Marks)

- A1. (a) Distinguish between discrete random variables and continuous random variables. [4 Marks]

- (b) A discrete random variable has probability mass function (p.m.f)

$$P(X = x) = \begin{cases} a(3 - x) & , \quad x = 0, 1, 2 \\ b & , \quad x = 3 \\ 0 & , \quad \text{otherwise} \end{cases}$$

Given that $E(X) = 1.6$, find

- (i) the values of a and b
(ii) $E(3X - 2)$ and show that $Var(X) = 1.64$ [7 Marks]

- A2. (a) The following table gives the Cumulative distribution function (c.d.f) of a discrete random variable, X .

X	1	2	4	5	6	8
$P(X \leq x)$	0.1	0.3	0.6	0.85	0.9	1.0

- (i) Derive the probability mass function (p.m.f) of X
(ii) Find $P(X > 5)$ and $P(3 < X \leq 6)$ [5 Marks]

- (b) Suppose that the random variable Y is equal to the number of hits obtained by a certain baseball player in his next 3 bats. Thus, Y takes on values 0, 1, 2, 3.

If $P(Y = 0) = 0.35$ and $P(Y = i) = 2iP(Y = i + 1)$ for $i = 1, 2$ find $E(Y)$.

[7 Marks]

- A3. The Ghana Health Service claims that only 20% of young adults can pass its physical fitness test. Suppose nine young adults are randomly selected and each is given the fitness test.

- (i) If W is the random variable that denotes the number of the nine young adults that past the test, what is the name given to W ? [2 Marks]
- (ii) What is the expected number of young adults who pass the test? [2 Marks]
- (iii) What is the probability that at least one of the young adults passes the test?

[3 Marks]

- A4. Suppose the random variable X has probability density function (p.d.f) given as

$$f(x) = \begin{cases} c(1-x^2) & , \quad -1 < x < 1 \\ 0 & , \quad \text{elsewhere} \end{cases}$$

for some constant c .

- (i) What is the value of c ? [4 Marks]
- (ii) Derive the cumulative distribution function of X , $F(X)$. [3 Marks]
- (iii) Hence or otherwise, find $P(X > 0.7)$. [3 Marks]

SECTION B (20 Marks)

ANSWER ONLY ONE QUESTION IN THIS SECTION

- B1. (a) The random variable X has the exponential distribution with probability density function

$$f(x) = \begin{cases} \theta e^{-\theta x} & , \quad x > 0, \theta > 0 \\ 0 & , \quad \text{Otherwise} \end{cases}$$

- (i) Show that expectation of X is $E(X) = \frac{1}{\theta}$ (3 Marks)
- (ii) Show that $P(X > a) = e^{-\theta a}$ for any $a > 0$. Deduce the median of X . (4 Marks)
- (iii) For any $b > 0$, find $P(X > a + b | X > a)$ and comment on this result. (4 Marks)

- (b) Now consider the case where $\theta = 1$.

- (i) Sketch the graph of $f(x)$. (4 Marks)
- (ii) State with a reason whether the distribution of X is positively or negatively skew. (2 Marks)
- (iii) Write down the mode of the distribution of X , and find the value of k such that $Mean - Mode = k[Mean - Median]$. (3 Marks)

- B2. (a) Write down the probability density function (pdf) of a continuous random variable X having the Normal distribution $N(\mu, \sigma^2)$ indicating the range of values of X and the parameter spaces for μ and σ^2 . (4 Marks)

- (b) State the properties of the Normal distribution. (2 marks)
- (c) Show that the random variable $Z = \frac{X - \mu}{\sigma} \sim N(0,1)$ and write down the pdf of Z . (4 Marks)
- (d) A student arrives at a school X minutes after 08:00 A.M., where X may be assumed to be normally distributed. On a particular day it is observed that 40 % of the students arrive before 08:30 A.M. and 90 % arrive before 08:55 A.M.
- (i) Find the mean and standard deviation of X . (7 Marks)
- (ii) The school has 1,200 students and classes start at 09:00 A.M. Estimate the number of students who will be late on that day. (3 Marks)